2023 Calculus II   Additional Exercise (6)

1. Exercise 8 in page 35 from Calculus Supplement Homework.

2. Exercise 5 in page 35 from Calculus Supplement Homework.

3. Exercise 18 in page 36 from Calculus Supplement Homework.

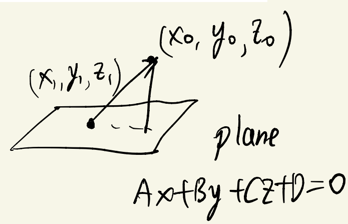
4. Suppose a line l passes through $P(1,2,1)$ and prependicular to the line $x = 1+3t$, $y = 2t$, $z = -1+t$. If l intersect with the line $x = 2s$, $y = s$, $z = -s$, find the equation of l.

5. Given the lines L_1: $x = 1+4t$, $y = 1+2t$, $z = -3+4t$ and L_2: $x = 3+2s$, $y = 2+s$, $z = -2+2s$. Can L_1 and L_2 determine only a plane M? If so, find the equation of M.

Q1 P35 Ex 8

8. Prove the distance between a point (x_0, y_0, z_0) to the plane $Ax + By + Cz + D = 0$ is given by

$$d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}.$$



解：单位法向量 $\vec{n} = \frac{1}{\sqrt{A^2+B^2+C^2}}(A, B, C)$

设 (x_1, y_1, z_1) 在平面上

$\vec{v} = (x_0 - x_1, y_0 - y_1, z_0 - z_1)$

$$d = \left||\vec{v}||\vec{n}|\cos\theta\right| = |\vec{v}\cdot\vec{n}| = \frac{1}{\sqrt{A^2+B^2+C^2}}\left|A(x_0-x_1)+B(y_0-y_1)+C(z_0-z_1)\right|$$

$$= \frac{|Ax_0+By_0+Cz_0+D|}{\sqrt{A^2+B^2+C^2}}$$

Q2 P35 Ex 5

5. Find the symmetry point of the point $(1, -2, 3)$ with respect to the plane $x+4y+z-14 = 0$.

Sol: normal vector $\vec{n} = (1,4,1)$

Assume symmetry point $Q: (1,-2,3) + \lambda(1,4,1)$

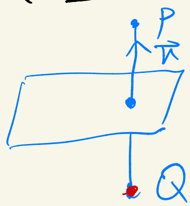
$= (1+\lambda, -2+4\lambda, 3+\lambda)$

of point $P: (1,-2,3)$

Then $\frac{1}{2}(P+Q) \in$ plane means

$(1+\frac{\lambda}{2}) + 4(-2+2\lambda) + (3+\frac{\lambda}{2}) - 14 = 0 \Longrightarrow$ Sol: $\lambda = 2$

$\therefore Q: (3, 6, 5)$

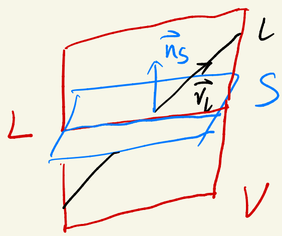


Q3 P36 Ex18

18. Give line $l: x = 1 + t, y = -1 - t, z = t$ and plane $S: 3x - y + z = 5$:

(a) Find the plane passing l and perpendicular to S;

(b) Find the equation of the projection of l to S.



Sol: (a) normal vector $\vec{n_S} = (3, -1, 1)$

direction of line l: $\vec{v_l} = (1, -1, 1)$

normal vector of plane V:

$$\vec{n_V} = \vec{n_S} \times \vec{v_l} = \left(\begin{vmatrix} -1 & 1 \\ -1 & 1 \end{vmatrix}, -\begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 3 & -1 \\ 1 & -1 \end{vmatrix} \right)$$

$$= (0, -2, -2)$$

$\therefore$ plane V: $-2(y-(-1)) - 2(z-0) = 0$

$$y + z + 1 = 0.$$

(b) L: projection of l to S 也就是 $V \cap S$:

$$\vec{V_L} = \vec{n_S} \times \vec{n_V} = \left(\begin{vmatrix} -1 & 1 \\ -2 & -2 \end{vmatrix}, -\begin{vmatrix} 3 & 1 \\ 0 & -2 \end{vmatrix}, \begin{vmatrix} 3 & -1 \\ 0 & -2 \end{vmatrix} \right) = (4, 6, -6)$$

$$\begin{cases} 3x - y + z - 5 = 0 \\ y + z + 1 = 0 \end{cases} \Longrightarrow P: (2, 0, -1) \in V \cap S$$

$$\therefore L: \begin{cases} x = 4t + 2 \\ y = 6t \\ z = -6t - 1 \end{cases}$$

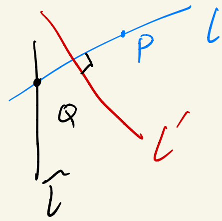
$\square$

Q4

4. Suppose a line l passes through $P(1,2,1)$ and prepedicular to the line $x = 1+3t$, $y = 2t$, $z = -1+t$. If l intersect with the line $x = 2s$, $y = s$, $z = -s$, find the equation of l.

Sol:

$$l' : \begin{cases} x = 1+3t \\ y = 2t \\ z = -1+t \end{cases} \qquad \tilde{l} : \begin{cases} x = 2s \\ y = s \\ z = -s \end{cases}$$



plane S prependicular to l' cotain P:

$$3(x-1) + 2(y-2) + (z-1) = 0$$

$$3x + 2y + z - 8 = 0$$

$S \cap \tilde{l} = \{Q\}$: $3 \cdot 2s + 2s - s - 8 = 0$

$$s = \frac{8}{7}$$

$\therefore Q : \left(\frac{16}{7}, \frac{8}{7}, -\frac{8}{7}\right)$

$$\overrightarrow{PQ} = \left(\frac{16}{7} - 1, \frac{8}{7} - 2, -\frac{8}{7} - 1\right) = \left(\frac{9}{7}, -\frac{6}{7}, -\frac{15}{7}\right)$$

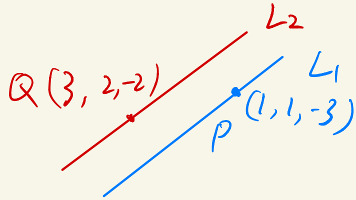
$$\therefore l : \begin{cases} x - 1 = 3t \\ y - 2 = -2t \\ z - 1 = -5t \end{cases}$$

□

Q5

5. Given the lines L_1: $x = 1+4t$, $y = 1+2t$, $z = -3+4t$ and L_2: $x = 3+2s$, $y = 2+s$, $z = -2+2s$. Can L_1 and L_2 determine only a plane M? If so, find the equation of M.

Sol:



$L_1 \parallel L_2$

$P: (1,1,-3) \in L_1$

$Q: (3,2,-2) \in L_2$

$\overrightarrow{PQ} = (2,1,1)$

$\vec{v}_{L_1} = (2,1,2)$

$$\vec{n} = \overrightarrow{PQ} \times \vec{v}_{L_1} = \left(\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}, -\begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix}, \begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} \right) = (1, -2, 0)$$

$\therefore M: \quad 1(x-1) - 2(y-1) = 0$

$x - 2y + 1 = 0$

12